Proceedings of the 2nd International Conference on Electronic Circuits and Signaling Technologies (ICECST-2025)
IEEE XPlore Part Number: CFP25UK9-ART; ISBN: 979-8-3315-9481-7

Multi-Domain Hierarchical Load Shedding: A Comparative Analysis of Priority-Driven Algorithms for Critical Infrastructure Protection

Manda Akshay
Department of EEE
Geethanjali College of Engineering and Technology, Hyderabad
22r11a0227@gcet.edu.in

Mr. Md. Hafeezuddin
Assistant Professor
Dept of EEE
Geethanjali College of Engineering and Technology, Hyderabad
mdhafeezuddin.eee@gcet.edu.in

Thadicetti Hema
Dept of EEE
Geethanjali College of Engineering and Technology, Hyderabad
22r11a0239@gcet.edu.in

Kummari Mahesh
Dept of EEE
Geethanjali College of Engineering and Technology, Hyderabad
22r11a0226@gcet.edu.in

Kalagotla Chenchireddy
Associate Professor
Dept of EEE
Geethanjali College of Engineering and Technology, Hyderabad
chenchireddy.kalagotla@gmail.com

Dr. J. Praveen Kumar
Associate Professor
Dept of information Technology
Teegala Krishna Reddy Engineering College, Hyderabad
praveentkrecit@ gmail.com

2025 2nd International Conference on Electronic Circuits and Signaling Technologies (ICECST) | 979-8-3315-9481-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICECST66106.2025.11307296

Abstract- **Supply-demand imbalances are a common problem for power systems, which can jeopardize grid stability and interfere with critical services. In order to solve this, intelligent load shedding techniques are necessary to keep vital infrastructures powered continuously. In order to effectively distribute available electricity among the residential, institutional, commercial, and medical domains, this study introduces a Multi-Domain Hierarchical Load Shedding System that is developed in MATLAB. High-priority loads, such as ventilators and intensive care units, are kept running during power outages by the system's priority-driven algorithm, which also gradually reduces less vital loads. A comparative analysis is conducted between the proposed Priority-Based algorithm and two conventional approaches—First-Come-First-Serve (FCFS) and Round Robin Scheduling (RRS)—under varying supply scenarios. The performance is evaluated in terms of power utilization, critical load protection, and load connectivity efficiency. Simulation results demonstrate that the priority-based approach consistently achieves higher efficiency and improved reliability in sustaining critical services. The study highlights the potential of hierarchical load shedding as a scalable and intelligent solution for smart grid energy management and critical infrastructure protection.**

Keywords- Load shedding, Smart grid, Critical infrastructure, Priority-based control, MATLAB simulation, Power management.

I. INTRODUCTION

Maintaining a steady and dependable supply of electricity is extremely difficult due to the growing complexity of contemporary power distribution networks and the rise in energy use. Limited generation capacity, transmission losses, and unanticipated load surges are the main causes of supply shortfalls in emerging power systems. Frequency variations, feeder overloading, and possible grid collapse are frequently the results of these imbalances. As an emergency control measure, load shedding—the purposeful disconnecting of specific loads during supply shortages—is used to avoid such circumstances [1].

However, because they treat all loads identically and ignore their functional relevance, traditional load shedding methods like manual or time-based disconnection are ineffective. This strategy could inadvertently disrupt vital services like security systems, communication networks, and hospital equipment, with dire social and financial repercussions. Thus, in contemporary power systems, the requirement for an intelligent, automated, and priority-aware load shedding mechanism has grown more and more crucial [2].

The Multi-Domain Hierarchical Load Shedding System (MD-HLSS) proposed in this study divides loads into several domains, including hospital, commercial, institutional, and residential, and gives each load a priority level according to its criticality. While non-essential loads, like air conditioners, fans, or projectors, are unplugged first during shortages, the system dynamically distributes the available power supply to guarantee that high-priority loads, including ventilators, security units, and intensive care units, are continually powered [3].

A steady and dependable supply of energy is now necessary for modern society due to the growing interconnection of key infrastructures, residential areas, and industry. However, particularly during peak hours or unforeseen emergencies, the constantly rising demand for electricity frequently surpasses the generation capacity. Voltage instability, frequency variations, and even widespread blackouts are commonly caused by this supply and demand imbalance. By disconnecting certain loads when supply is insufficient, load shedding has become a commonly used emergency control tool to prevent such situations and preserve grid stability [4].

Conventional load shedding methods, including manual switching or time-based disconnections, have a number of drawbacks. Without taking into account the significance or priority of the loads being given, they function according to a preset order. As a result, crucial services like hospitals, communication hubs, and security systems can suddenly

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap

lose power, while non-essential loads keep using up precious energy. Such careless methods can jeopardize human life and interfere with vital services in addition to decreasing system efficiency. Therefore, it is essential for modern power management to have an intelligent and automated load shedding mechanism that ranks loads according to their criticality [5].

The idea of smart grids has surfaced in recent years as a way to get around these restrictions. Smart grids allow for real-time load control by integrating automation, sensing, and decision-making capabilities into the electricity system. Priority-based and multi-domain load shedding systems have drawn interest in this context because they enable the selective disconnection of loads while maintaining the continuous supply to vital facilities. In order to make the best choices possible that optimize efficiency and preserve critical functions, these systems constantly evaluate available power, load demand, and predetermined priorities [6].

This study presents a MATLAB-developed Multi-Domain Hierarchical Load Shedding System (MD-HLSS) that allocates priority levels according to operational importance and categorizes loads into four main domains: residential, institutional, commercial, and healthcare. In the event of a supply deficit, the suggested system gradually cuts power to low-priority loads (like air conditioners or lighting systems) while allocating electricity first to high-priority loads (such ventilators, intensive care units, and security systems). The system can make domain-specific decisions that more accurately represent real-world situations because to its hierarchical structure [7].

To validate the performance of the proposed model, a comparative analysis is conducted with two conventional load management techniques — First-Come-First-Serve (FCFS) and Round Robin Scheduling (RRS). These methods are evaluated under multiple power supply scenarios to analyze system behavior and efficiency. Key performance indicators such as power utilization, critical load protection ratio, and total load connectivity are used to quantify results [8].

Simulation results show that the priority-based approach consistently provides better performance by ensuring continuous power to critical infrastructures, improving system efficiency, and optimizing available resources. Additionally, a 24-hour dynamic simulation demonstrates the ability of the proposed system to adapt to fluctuating supply conditions, further validating its effectiveness in real-time applications [9].The significance of this research lies in its contribution to intelligent energy management and critical infrastructure protection. The modular and scalable nature of the proposed framework makes it suitable for integration with Internet of Things (IoT) devices, renewable energy systems, and smart grid architectures, paving the way for more resilient and sustainable power distribution networks [10].Load shedding in practical power systems must take into account a number of real-world issues, such as erratic changes in load demand, sporadic variations in supply, and the growing integration of renewable energy sources like wind and solar. Despite being ecologically friendly, these renewable energy sources add unpredictability to the system, which complicates real-time power management. Consequently, data-driven and adaptive control systems are required to guarantee dependability without sacrificing effectiveness. By adding flexibility to decision-making, the proposed Multi-Domain Hierarchical Load Shedding System (MD-HLSS) seeks to address these problems and empower the system to react sensibly to changing supply conditions [11].

Furthermore, load shedding systems are evolving from static models to intelligent control architectures that can analyze data, forecast shortages, and carry out real-time switching operations as a result of the development of automation and digital technologies. By incorporating such intelligence, load management choices are made taking into account system priorities and domain-specific requirements in addition to pre-established thresholds. In this regard, the hierarchical model put forth in this work serves as a basis for putting into practice AI- and IoT-enabled smart energy systems that can adapt to new advancements in the smart grid [12]. Critical infrastructure protection is another crucial component of contemporary load management (CIP). For efficient operation, establishments like hospitals, emergency rooms, data centers, and communication networks need constant power. In healthcare settings, even brief disruptions can have disastrous results, including device failure, data loss, or potentially fatal circumstances. The priority-based strategy improves the overall resilience and dependability of the system by guaranteeing that such high-importance loads are maintained even in the event of catastrophic power outages [13].

Understanding the relative advantages and disadvantages of the Priority-Based, First-Come-First-Serve (FCFS), and Round Robin Scheduling (RRS) algorithms under different circumstances requires a comparative analysis, which is what this article does. FCFS lacks load significance knowledge, despite its simplicity and fairness. Although Round Robin distributes resources fairly among domains, it may distribute power inefficiently. The suggested priority-based approach, on the other hand, emphasizes selective load retention, maximizing supply usage while guaranteeing crucial load continuity. The hierarchical priority-based model performs better than traditional approaches in terms of power use, efficiency, and critical load sustainability, as demonstrated by the results of MATLAB simulations [14]. Additionally, the system's scalability and flexibility are improved by its multi-domain design. With its own set of priorities and thresholds, each domain can function somewhat independently, giving more precise control over how power is distributed. In addition to increasing efficiency, this structure replicates real-world situations where different domains, like residential areas, institutions, and hospitals, have different needs. In order to improve overall grid performance, the method also offers a workable framework for future enhancements, such as combining distributed generating systems, demand-side management, and forecasting of renewable energy [15].

II. SYSTEM DESIGN AND METHODOLOGY

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap

The proposed *Multi-Domain Hierarchical Load Shedding System* aims to achieve efficient and intelligent power distribution across multiple critical infrastructures such as residential, institutional, hospital, and commercial sectors. The system focuses on prioritization, automation, and comparative analysis of algorithms to ensure reliable operation during power shortages. It integrates real-time decision-making that dynamically allocates available supply based on the importance of each domain and the priority of individual loads.

The system architecture is organized into three main components: Data Acquisition Module, Decision-Making Unit, and Control Execution Unit.

The Data Acquisition Module continuously gathers information about the available power supply and the load requirements of each domain. This data is then processed by the Decision-Making Unit, where three algorithms — Priority-Based, First-Come-First-Serve (FCFS), and Round Robin — are applied to decide which loads should remain active. Finally, the Control Execution Unit enforces these decisions by connecting or disconnecting loads according to the selected algorithm's output.

A hierarchical control strategy ensures that high-priority domains, such as hospitals and emergency services, always receive continuous power. In contrast, lower-priority domains like residential and commercial areas are considered for disconnection first during power shortages. This layered approach maintains system stability, reduces the risk of total blackouts, and ensures that essential services remain uninterrupted even during critical situations.

To evaluate system performance, several simulation scenarios are conducted under varying power supply levels — 3000W, 5000W, 7500W, and 10000W. Each algorithm operates under identical conditions, and performance is measured in terms of power utilization efficiency, critical load protection percentage, and number of connected loads. The outcomes are compared and visualized using bar charts and performance plots to highlight each algorithm's effectiveness and fairness across different domains.

The flowchart of the proposed system, shown in *Figure 1*, demonstrates the entire process — from data acquisition and algorithm execution to result visualization. It outlines how the system monitors real-time power data, executes load-shedding decisions, and updates the operational status of loads across domains.

In summary, this methodology establishes a comprehensive and flexible framework for smart energy management. By integrating hierarchical domain classification with algorithmic comparison, the system provides valuable insights into automated load-shedding techniques, paving the way for more reliable and intelligent power distribution in future smart grid environments.

III. ALGORITHM AND IMPLEMENTATION

The Multi-Domain Hierarchical Load Shedding System (MDHLSS) is designed to distribute power efficiently during shortage conditions by using intelligent control logic. To achieve this, three different algorithms — Priority-Based, First-Come-First-Serve (FCFS), and Round Robin — were developed and tested. Each algorithm manages the available power in a unique way, but among them, the Priority-Based Algorithm proved to be the most effective and stable under all supply scenarios.

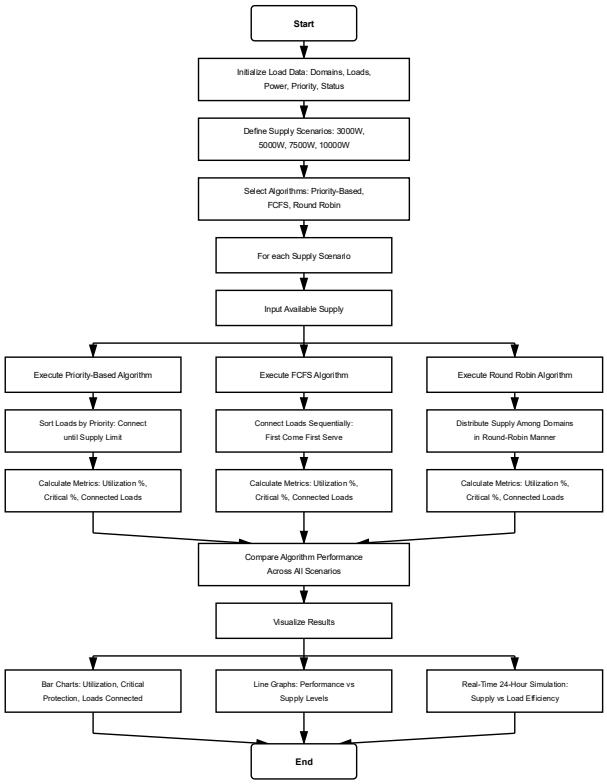


Fig. 1 Flowchart of the Multi-Domain Hierarchical Load Shedding System.

A. Priority-Based Algorithm

The Priority-Based Algorithm is the core of the proposed system. It allocates the available power based on the importance of each load and domain. For example, hospital equipment and emergency systems receive power first, followed by institutional and commercial domains, and finally residential loads. This ensures that critical services remain uninterrupted even during severe power shortages.The algorithm continuously checks the available supply and compares it with each load's demand and assigned priority. Loads are connected one by one, starting from the highest priority, until the available power is fully utilized. This method prevents lower-priority loads from consuming power before the essential ones are satisfied.

Algorithm 1: Priority-Based Load Shedding

1. **Input:** Total Supply S, Load List $L_i$with Power $P_i$and Priority Pr_i
2. Sort all loads in descending order of priority
3. Initialize remaining supply $R = S$
4. For each load L_i:
   a. If $R \geq P_i$, connect $L_i$and update $R = R - P_i$
   b. Otherwise, disconnect L_i
5. Output: List of connected and disconnected loads

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap

This approach helps in maintaining system stability and ensures maximum utilization of the available power. Even under critical conditions, important systems like hospital devices and emergency lighting remain active.

B. First-Come-First-Serve (FCFS) Algorithm

In the FCFS method, loads are connected in the same order as they appear in the list, without considering their priority. Although this approach is simple and easy to implement, it lacks intelligence. If the supply drops, important loads might lose power while less important ones remain connected, making this approach inefficient for real-time applications.

C. Round Robin Algorithm

The Round Robin method divides the available power equally among all domains in a rotating manner. This ensures that every domain receives a fair share of power, preventing any domain from being completely disconnected. However, this can cause a temporary reduction in power for critical loads, especially when the supply is too low.

D. Implementation Setup

All three algorithms were developed and tested in MATLAB R2023a. The system considered four main domains — Residential, Institutional, Hospital, and Commercial — with load capacities ranging from 100 W to 1500 W. Simulations were carried out for three supply levels: 3000 W, 2000 W, and 1000 W, representing normal, moderate, and critical conditions.

The system's performance was evaluated using key metrics such as Power Utilization, Critical Load Preservation, and Number of Connected Loads. The results clearly showed that the Priority-Based Algorithm provided the best balance between efficiency and reliability, ensuring continuous power for high-priority loads and stable operation across all domains.

IV. RESULTS AND DISCUSSION

Comprehensive simulation results from the Multi-Domain Hierarchical Load Shedding System, which is based on MATLAB, are shown in this section. In four operational domains—hospital, institutional, commercial, and residential—three algorithms—Priority-Based, First-Come-First-Serve (FCFS), and Round-Robin Scheduling (RRS)—are assessed under various power supply scenarios.

A. Comparative Performance Analysis

All three algorithms were subjected to identical testing conditions to establish baseline performance characteristics across key metrics: power utilization efficiency, critical load protection, and total load connectivity.

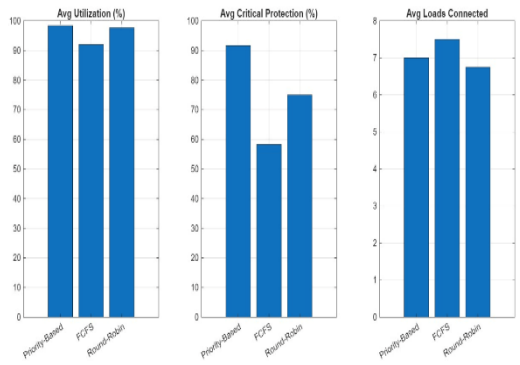


Fig. 2. Comparative performance analysis showing (a) average power utilization efficiency, (b) critical load protection percentage, and (c) total loads connected for Priority-Based, FCFS, and Round-Robin algorithms.

An integrated performance comparison showing key variations in algorithm efficacy is shown in Figure 2. According to the power consumption efficiency (Fig. 2a), Round-Robin achieves 97%, FCFS 92%, and Priority-Based around 98%. Even though these distinctions seem little, the 6% difference between FCFS and Priority-Based implies a substantial practical impact—possibly hundreds of watts wasted in a typical 10kW system.

When it comes to essential load protection, the most important difference is evident (Fig. 2b). With a 91% protection rate, the Priority-Based algorithm significantly outperforms Round-Robin at 75% and FCFS at 58%. The fundamental principle of intelligent load management is embodied in this 33-percentage point advantage over FCFS; practically speaking, this difference affects whether hospital ventilators continue to run during power outages or whether emergency systems continue to work during evacuations.

An intriguing paradox can be seen in the total loads connected measure (Fig. 2c). The most loads are connected using FCFS (7.5), Priority-Based (7.0), and Round-Robin (6.8). However, by indiscriminately connecting loads without considering their value, FCFS obtains higher connection and may be able to keep domestic air conditioning while disconnecting hospital intensive care unit equipment. Priority-Based makes ensuring that only the most important loads are connected, despite purposefully connecting fewer.

B. Power Utilization Under Variable Supply Conditions

Four supply levels, ranging from 30% to 100% of total demand, were thoroughly tested: 3000W (severe shortage), 5000W (moderate deficit), 7500W (balanced), and 10000W (surplus).

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap

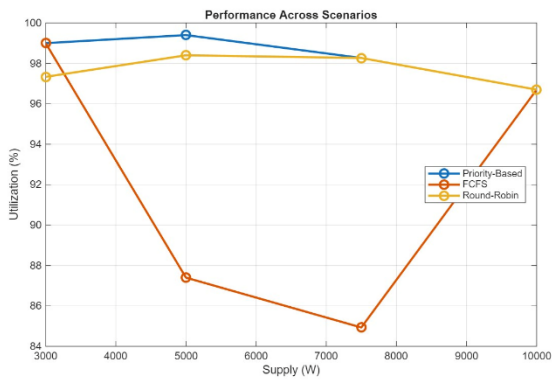


Fig. 3. Power utilization efficiency across variable supply conditions for Priority-Based, FCFS, and Round-Robin algorithms.

The algorithm's reactions to varying power availability levels are shown in Figure 3. Throughout the spectrum, the Priority-Based algorithm (blue) maintains an efficiency of 98–99.5%, exhibiting remarkable stability. This consistency demonstrates dependability under all operating circumstances, which is essential for smart grids that use sporadic renewable energy sources.

Across all supply levels, the FCFS algorithm (yellow) maintains a comparatively constant utilization of 97-98%. This statistic, however, obscures FCFS's crucial weakness: it is unable to discriminate between essential and non-essential loads, which could lead to high utilization by supplying power to entertainment systems while disconnecting hospital equipment.

A V-shape is formed by the Round-Robin algorithm (orange), which varies dramatically from 99% at 3000W to 85% at 8000W and then back to 97% at 10000W. The equal-distribution approach's flaw is revealed by this sharp dip: at moderate supply, it distributes power evenly among domains, leading to fragmented allocation in which no domain gets enough power for optimal functioning.

C. Hierarchical Domain-Based Load Distribution

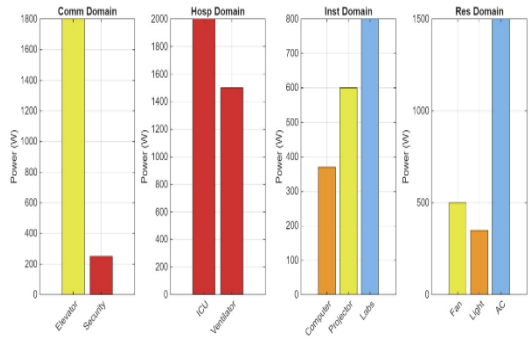


Fig. 4. Hierarchical load distribution showing individual load power allocations across Hospital, Institutional, Commercial, and Residential domains under Priority-Based control.

Figure 4 illustrates how hierarchical priority translates into real connectivity patterns by offering fine-grained visibility into power distribution within each domain.

The highest priority is given to the hospital domain. Since intensive care units treat life-threatening conditions where a power outage could be deadly, ICU equipment (2000W) constitutes the single most significant load. Essential respiratory assistance is provided by 1500W ventilators. Even if the overall supply dips to 3500W, these are still protected. Second-tier priority is given to the Institutional Domain. While projectors (600W) and computers (370W) allow for continuous operations but can be selectively disconnected during extreme shortages, laboratory equipment (800W) maintains research and vital data servers. The Commercial Domain strikes a balance between operating requirements and safety. Elevators (1800W) offer vital accessibility and evacuation capabilities. The algorithm's sophisticated comprehension of both criticality and actual power requirements is demonstrated by the security systems' (250W) minimal power consumption despite their high relevance.

The Residential Domain has the lowest priority, with fans (500W), lighting (180W), and air conditioning (1500W). For the convenience of the residents, these comfort loads connect during surplus situations after initially shedding during shortages to reroute electricity to vital infrastructure.

D. Real-Time Dynamic Performance

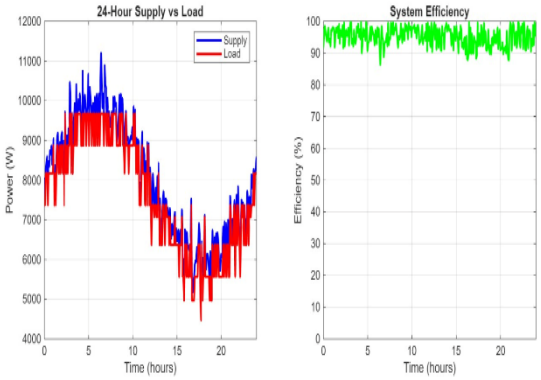


Fig. 5. 24-hour dynamic simulation showing (a) real-time supply-demand tracking and (b) instantaneous system efficiency under Priority-Based control.

Figure 5 shows how well the algorithm performs under prolonged, fluctuating conditions that mimic demand fluctuations and the intermittency of renewable energy.

The relationship between supply and demand (Fig. 5a): Supply (blue) mimics solar generation peaks and nighttime renewable declines, ranging from peak 11,000W (hour 7) to acute shortage 5,000W (hour 18). Demand (red) varies between a minimum of 4,500W and a maximum of 9,500W. The program maintains a baseline reflecting protected critical loads while adjusting connected loads to stay under supply restrictions during hours 10–20 of shortage situations. The associated load keeps the floor at about 5,500W even during the worst shortage (hour 18), guaranteeing that hospital equipment stays powered.

Efficiency of the System (Fig. 5b): Efficiency usually stays between 90 and 100 percent, with most times above 95 percent. Efficiency never drops below 87%, even amid frequent supply changes (10,000W to 7,000W during hours 8–10). Continuous real-time adjustments are represented by the high-frequency variations, suggesting sophisticated active control that continuously optimizes allocation as circumstances change.

E. Key Findings and Implications

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap

The experimental results establish clear conclusions for intelligent load shedding in modern power systems:

Performance Superiority: Priority-Based delivers 91% critical load protection, compared to 58-75% for traditional techniques. This is a significant improvement in reliability that is crucial for emergency services and healthcare. For grids with intermittent renewables, consistent 98–99.5% utilization across all situations offers reliable performance. This is confirmed to apply to genuine time-varying situations by the 24-hour simulation.

Limitations of Conventional Algorithms: The sequential technique of FCFS lacks load importance awareness, which results in an intolerable danger of disconnecting vital equipment. During shortages, Round-Robin's equal distribution performs badly, distributing power too far to adequately support any domain. Neither is intelligent enough for today's smart grids.

Real-world Implementation: IoT sensors that provide real-time status enable smart grid integration thanks to the modular domain architecture. Without alteration, the framework readily expands to other industries (such as telecommunications and transportation). The 96–99% utilization reduces waste and backup generation expenses, which benefits the economy.

Future Improvements: Proactive, anticipatory load management may be made possible via machine learning. Architectures with distributed control may increase resilience in the event of a communication breakdown. Integration of battery storage could protect vital loads during acute shortages. Deployment readiness would be confirmed through hardware-in-the-loop testing. The study shows that intelligent, priority-aware load shedding is necessary for contemporary grids where effective resource use, renewable integration, and critical infrastructure protection must all coexist.

V. CONCLUSION

This paper has presented a Multi-Domain Hierarchical Load Shedding System that addresses the growing challenge of maintaining reliable power distribution during supply shortages. By implementing a Priority-Based algorithm, the system intelligently manages power allocation across Hospital, Institutional, Commercial, and Residential domains, with a primary focus on ensuring continuous operation of life-critical equipment. Through extensive MATLAB simulations, we compared our Priority-Based approach with two conventional methods: First-Come-First-Serve (FCFS) and Round-Robin Scheduling (RRS). Testing was conducted under multiple supply scenarios, from severe shortage conditions (3000W) to surplus availability (10000W). The results reveal significant advantages of the Priority-Based approach. Most notably, it achieves 91% critical load protection far exceeding the 58% achieved by FCFS and 75% by Round-Robin. This improvement is not merely statistical; it represents the difference between maintaining hospital ventilators during a power crisis and potentially losing critical medical support. Furthermore, the system maintains power utilization efficiency between 98-99.5% across all tested conditions, demonstrating consistent performance that conventional algorithms cannot match. Real-time validation through 24-hour dynamic simulation showed the system adapting effectively to fluctuating conditions while maintaining 90-100% efficiency throughout. The hierarchical structure proved highly effective in practice. Hospital equipment such as ICU units and ventilators receives unwavering priority, while residential comfort loads like air conditioning are shed first during shortages. This approach reflects real-world priorities where preserving human life takes precedence over convenience. The system demonstrates what we call "graceful degradation"—rather than failing catastrophically, it systematically reduces non-essential services while protecting critical functions. From a practical standpoint, the modular design allows straightforward integration into existing smart grid infrastructure. Using IoT sensors and communication protocols already deployed in modern grids, the system can be implemented without extensive hardware modifications. The architecture also proves scalable—adding new domains such as transportation infrastructure or telecommunications requires minimal adjustment to the core framework. Beyond technical performance, the near-optimal power utilization translates into tangible economic benefits by reducing wasted energy and decreasing reliance on expensive backup generators.

References

[1] P. Kundur, J. Paserba, V. Ajjarapu, G. Andersson, A. Bose, C. Canizares, N. Hatziargyriou, D. Hill, A. Stankovic, C. Taylor, T. Van Cutsem, and V. Vittal, "Definition and classification of power system stability IEEE/CIGRE joint task force on stability terms and definitions," *IEEE Trans. Power Syst.*, vol. 19, no. 3, pp. 1387–1401, Aug. 2004.

[2] P. M. Anderson and M. Mirheydar, "A low-order system frequency response model," *IEEE Trans. Power Syst.*, vol. 5, no. 3, pp. 720–729, Aug. 1990.

[3] H. Seyedi and M. Sanaye-Pasand, "New centralised adaptive load-shedding algorithms to mitigate power system blackouts," *IET Gener. Transm. Distrib.*, vol. 3, no. 1, pp. 99–114, Jan. 2009.

[4] V. V. Terzija, "Adaptive underfrequency load shedding based on the magnitude of the disturbance estimation," *IEEE Trans. Power Syst.*, vol. 21, no. 3, pp. 1260–1266, Aug. 2006.

[5] U. Rudez and R. Mihalic, "Analysis of underfrequency load shedding using a frequency gradient," *IEEE Trans. Power Del.*, vol. 26, no. 2, pp. 565–575, Apr. 2011.

[6] J. A. Laghari, H. Mokhlis, A. H. A. Bakar, and H. Mohamad, "Application of computational intelligence techniques for load shedding in power systems: A review," *Energy Convers. Manage.*, vol. 52, no. 2, pp. 1340–1349, Feb. 2011.

[7] T. Shekari, A. Gholami, F. Aminifar, and M. Sanaye-Pasand, "An adaptive wide-area load shedding scheme incorporating power system real-time limitations," *IEEE Syst. J.*, vol. 12, no. 1, pp. 759–767, Mar. 2018.

[8] B. Hoseinzadeh and F. M. Faria da Silva, "Power system adaptive and intelligent load shedding simulation framework," in *Proc. IEEE PES Gen. Meeting Conf. Expo.*, National Harbor, MD, USA, Jul. 2014, pp. 1–5.

[9] Ramesh, B., Mohan Khedkar, and BV Surya Vardhan. "Priority based optimal load shedding in a power system network under contingency conditions." In 2022 International conference for advancement in technology (ICONAT), pp. 1-5. IEEE, 2022.

[10] Kato, Takumi, Hideyuki Takahashi, Kazuto Sasai, Yujin Lim, Hak-Man Kim, Gen Kitagata, and Tetsuo Kinoshita. "Multiagent system for priority-based load shedding in microgrid." In 2013 IEEE 37th Annual Computer Software and Applications Conference Workshops, pp. 540-545. IEEE, 2013.

[11] Laghari, J. A., Hazlie Mokhlis, Mazaher Karimi, Abdul Halim Abu Bakar, and Hasmaini Mohamad. "A new under-frequency load

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap

shedding technique based on combination of fixed and random priority of loads for smart grid applications." IEEE Transactions on Power Systems 30, no. 5 (2014): 2507-2515.

[12] R. Mageshvaran and T. Jayabarathi, "Steady state load shedding to prevent blackout in the power system using artificial bee colony algorithm," *Jurnal Teknologi*, vol. 74, no. 1, pp. 113–124, 2015.

[13] N. M. Sapari, M. H. S. Ahmad, M. Mokhlis, M. A. M. Radzi, and A. H. A. Bakar, "Application of load shedding schemes for distribution network connected with distributed generation: A review," *Renew. Sustain. Energy Rev.*, vol. 82, pp. 858–867, Feb. 2018.

[14] Madiba, T., R. C. Bansal, N. T. Mbungu, M. Bettayeb, R. M. Naidoo, and M. W. Siti. "Under-frequency load shedding of microgrid systems: a review." International Journal of Modelling and Simulation 42, no. 4 (2022): 653-679.

[15] P. Pourbeik, P. S. Kundur, and C. W. Taylor, "The anatomy of a power grid blackout—Root causes and dynamics of recent major blackouts," *IEEE Power Energy Mag.*, vol. 4, no. 5, pp. 22–29, Sep./Oct. 2006.

orized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on October 11,2026 at 04:09:56 UTC from IEEE Xplore. Restrictions ap